
Remark: Each problem mentions which lectures it uses techniques from. If you would like feedback on your homework, you can either submit it in class or email it to both of us. If you find any errors, please let us know!

Recap

For arbitrary sets A and B , we use the following notation:

1. We write $|A| = |B|$ if there exists a bijection $f : A \rightarrow B$. Equivalently, $|A| = |B|$ if and only if there exists a one-one function $f : A \rightarrow B$ and a one-one function $g : B \rightarrow A$.
2. We say $|A| \leq |B|$ if there exists a one-one function from A to B .
3. We say $|A| < |B|$ if $|A| \leq |B|$ and $|A| \neq |B|$. Equivalently, $|A| \leq |B|$ and there is no one-one function from B to A .
4. A set A is *countable* if $|A| \leq |\mathbb{N}|$. It is *countably infinite* if $|A| = |\mathbb{N}|$.

We use $\mathcal{P}(X) = \{A : A \subseteq X\}$ to denote the power set of X . We also write $\{0, 1\}^*$ for the set of all finite binary strings, including the empty string ε .

1 Lectures 1 and 2

1. In each of the following questions, choose either true or false.
 - (a) If $A \subseteq B$, then $|A| \leq |B|$.
☐ True
☐ False
 - (b) If $A \subsetneq B$, then $|A| < |B|$.
☐ True
☐ False
 - (c) Every subset of a countable set is countable.
☐ True
☐ False
 - (d) Every subset of a countably infinite set is countably infinite.
☐ True
☐ False
 - (e) Every language $L \subseteq \{0, 1\}^*$ is countable.
☐ True
☐ False
 - (f) The set of all languages $\mathcal{L} = \{L : L \subseteq \{0, 1\}^*\}$ is countable.
☐ True
☐ False

2. Let $\mathcal{S} = \bigcup_{k \in \mathbb{N}} \mathbb{N}^k$ be the set of all finite sequences of non-negative integers, where $\mathbb{N} = \{0, 1, 2, \dots\}$. The case $k = 0$ corresponds to the empty sequence. Show that $\mathcal{S}$ is countable using one of the following methods. You may use the fact that $\{0, 1\}^*$ is countable.

1. Construct a one-one function

$$f : \mathcal{S} \longrightarrow \{0, 1\}^*.$$

Make sure to explain why your function is one-one.

2. Give an enumeration of $\mathcal{S}$ in which every element has a finite position.

Hint. For example, you may organize the sequences according to the value

$$k + n_1 + \dots + n_k$$

for a sequence $(n_1, \dots, n_k) \in \mathbb{N}^k$.

3. For an arbitrary set X , let $\mathcal{P}(X) = \{A : A \subseteq X\}$ denote the power set of X . Prove the following.

(a) Show that $|\mathbb{N}| < |\mathcal{P}(\mathbb{N})|$.

(b) Prove that there is no bijection $f : \mathcal{P}(X) \longrightarrow X$. Conclude that $|\mathcal{P}(X)| > |X|$.

Hint: Prove by contradiction. Assume f is a bijection. Define $D_f \in \mathcal{P}(X)$ by $D_f = \{x \in X \mid x \notin f^{-1}(x)\}$.